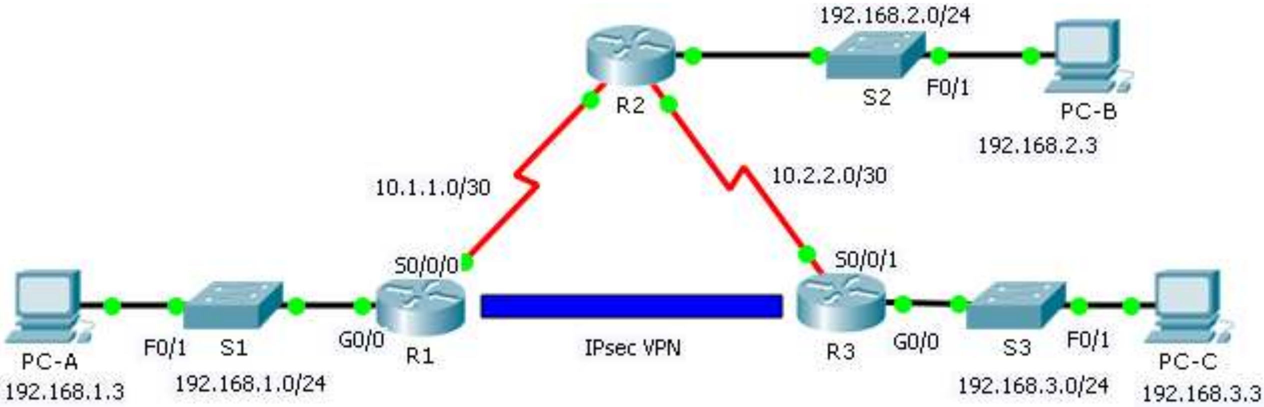
Configuring Site-to-Site VPNs

Introduction

In this lab, I learned how to configure a site-to-site IPsec VPN using Cisco Packet Tracer to securely connect two separate LANs over an untrusted network. The lab focused on enabling security features on Cisco routers, creating access control lists to identify interesting traffic, and configuring ISAKMP Phase 1 and IPsec Phase 2 settings. I also explored how to apply crypto maps and verify tunnel establishment through packet encryption and decryption, simulating secure communication between two remote networks.

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	10.1.1.2	255.255.255.252	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A
	S0/0/0	10.1.1.1	255.255.255.252	N/A
	S0/0/1	10.2.2.1	255.255.255.252	N/A
R3	G0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/1	10.2.2.2	255.255.255.252	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1

PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1

ISAKMP Phase 1 Policy Parameters

Parameters		R1	R3
Key distribution method	Manual or **ISAKMP**	ISAKMP	ISAKMP
Encryption algorithm	**DES**, 3DES, or AES	AES	AES
Hash algorithm	MD5 or **SHA-1**	SHA-1	SHA-1
Authentication method	Pre-shared keys or **RSA**	pre-share	pre-share
Key exchange	DH Group **1**, 2, or 5	DH 2	DH 2
IKE SA Lifetime	86400 seconds or less	86400	86400
ISAKMP Key		cisco	cisco

Bolded parameters are defaults. Other parameters need to be explicitly configured.

IPsec Phase 2 Policy Parameters

Parameters	R1	R3
Transform Set	VPN-SET	VPN-SET
Peer Hostname	R3	R1
Peer IP Address	10.2.2.2	10.1.1.2
Network to be encrypted	192.168.1.0/24	192.168.3.0/24
Crypto Map name	VPN-MAP	VPN-MAP
SA Establishment	ipsec-isakmp	ipsec-isakmp

Objectives

Part 1: Enable Security Features

Part 2: Configure IPsec Parameters on R1

Part 3: Configure IPsec Parameters on R3

Part 4: Verify the IPsec VPN

Scenario

In this activity, I will configure two routers to support a site-to-site IPsec VPN for traffic flowing from their respective LANs. The IPsec VPN traffic will pass through another router that has no knowledge of the VPN. IPsec provides secure transmission of sensitive information over unprotected networks such as the Internet. IPsec acts at the network layer, protecting and authenticating IP packets between participating IPsec devices (peers), such as Cisco routers

Part 1: Enable Security Features

In this section, I learned how to enable security features on Cisco routers using the securityk9 module.

Confirming configurations:

R1

CLI

IOS Command Line Interface

```
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#interface s0/0/0
R1(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/1     192.168.1.1     YES NVRAM  up                    up
Serial0/0/0            10.1.1.2        YES NVRAM  up                    up
Serial0/0/1            unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
R1#
```

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Top

Noting that R1 configurations were not according to the addressing table I disconnected the cable from G0/1 to G0/0 then ran the following commands to correct the mismatch

R1

CLI

IOS Command Line Interface

```
R1(config)#
R1(config)#interface G0/1
R1(config-if)#no ip address
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

R1(config-if)#interface G0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#
```

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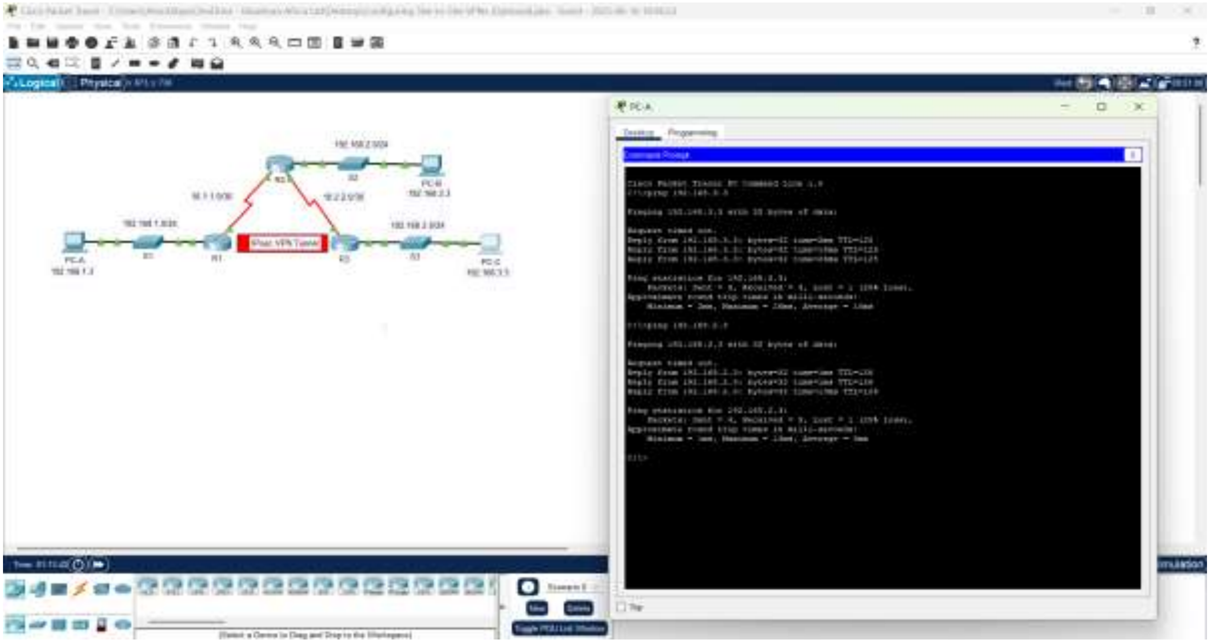
The commands start by removing the ip addressing from GO/1 and shuting it down then assigning the ip addressing to the correct interface which according to the table isG0/0.

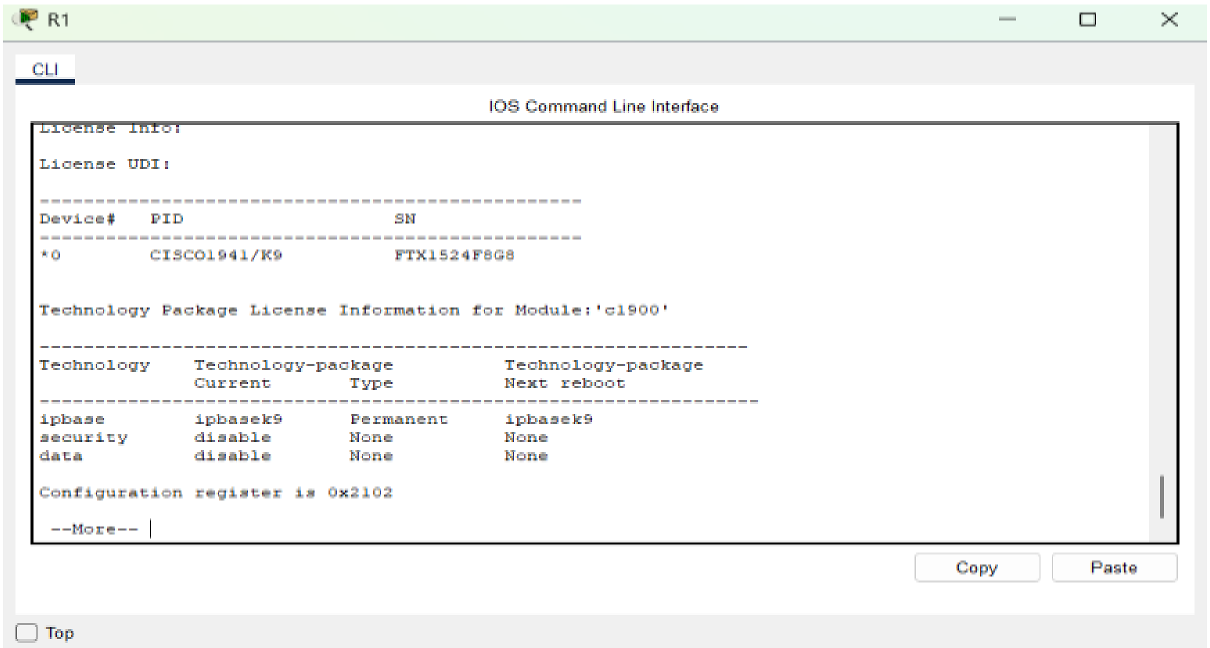
Step1: Test Connectivity

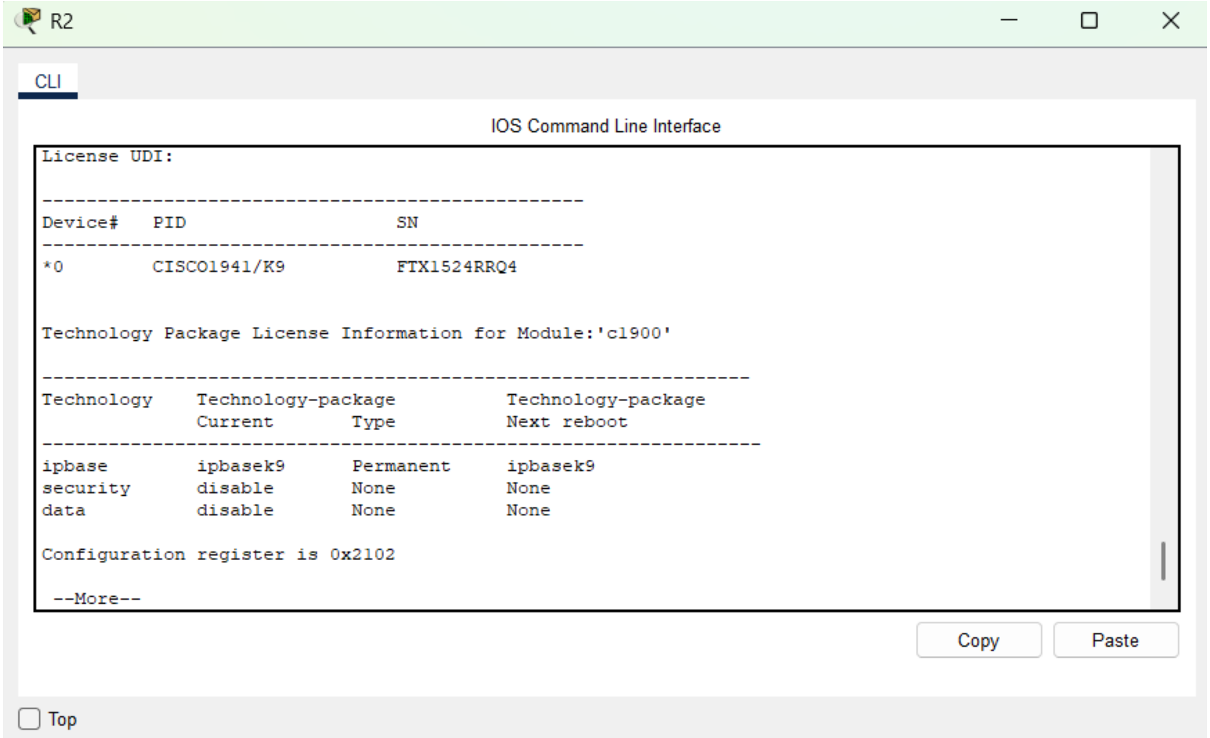
I started by testing connectivity where from PCA pinged PCB and PCC which was a success.

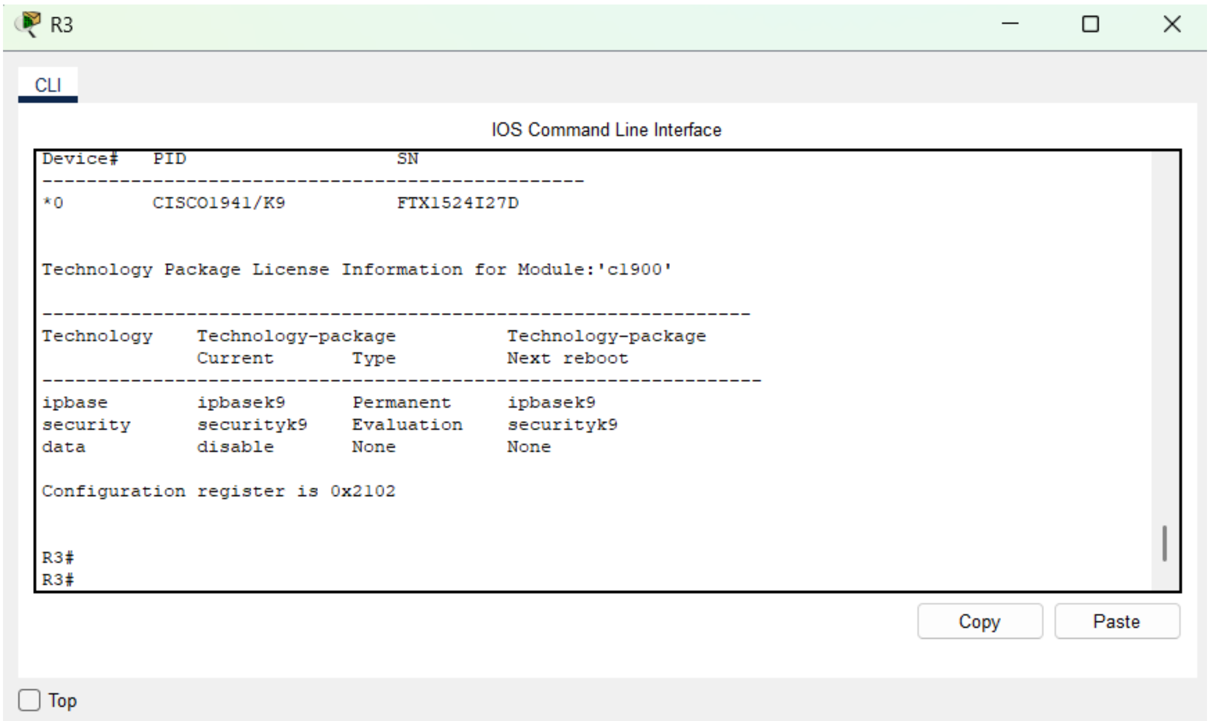


Step 2: Enable the Security Technology Package

1. **Verified the license status** using the show version command to check if the security technology package was active.







From the 3 screenshots, only one R3 has the security technology active. Hence the next step is to activate the technology in R1 and R2.

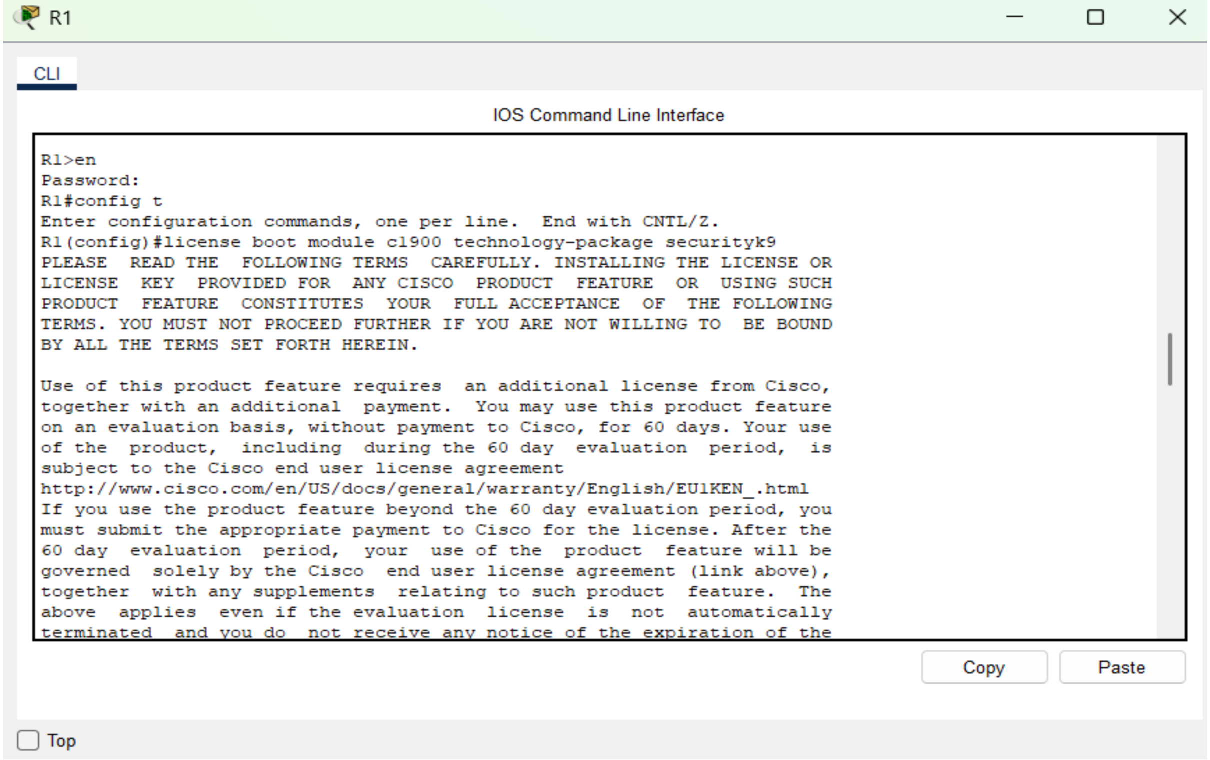
2. **Activated the securityk9 module** if it was not enabled:

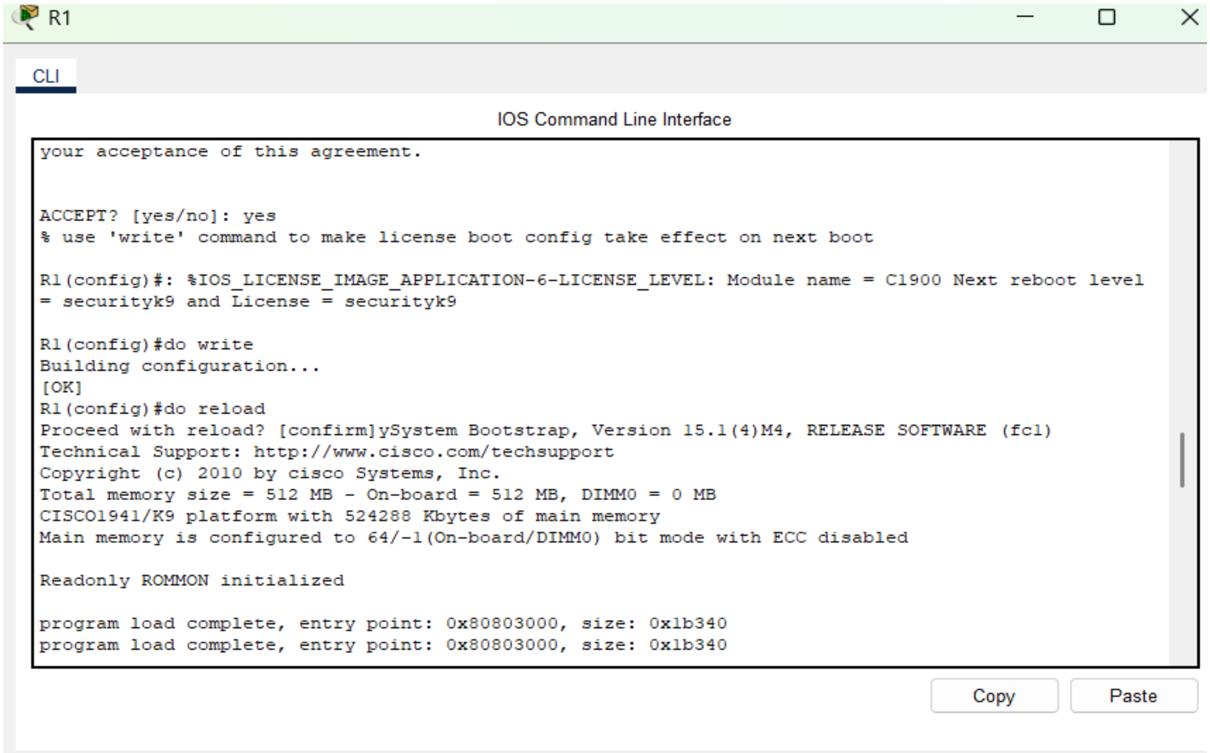
R1(config)# license boot module c1900 technology-package securityk9

R1(config)# end

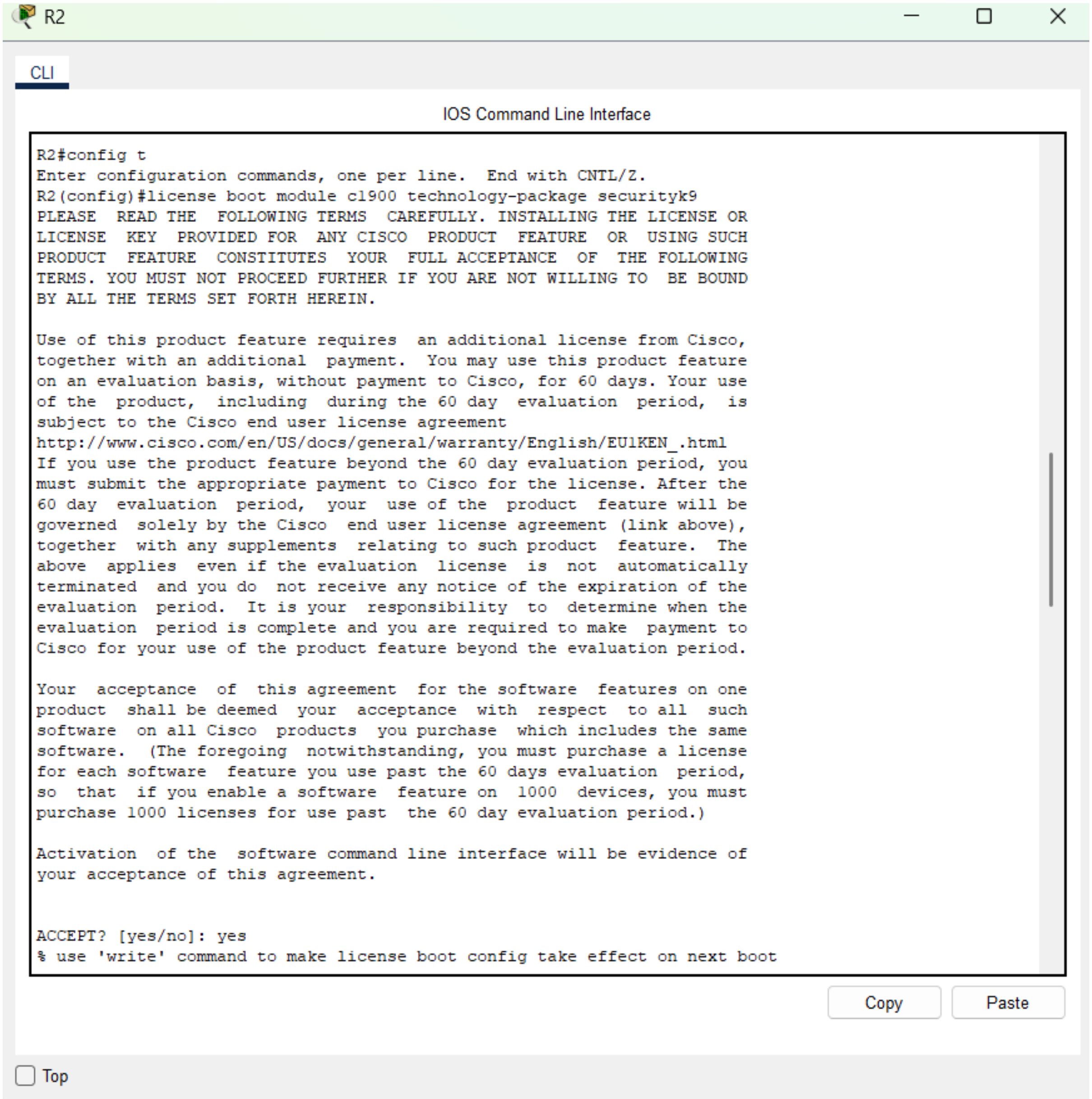
R1# do write

R1# reload

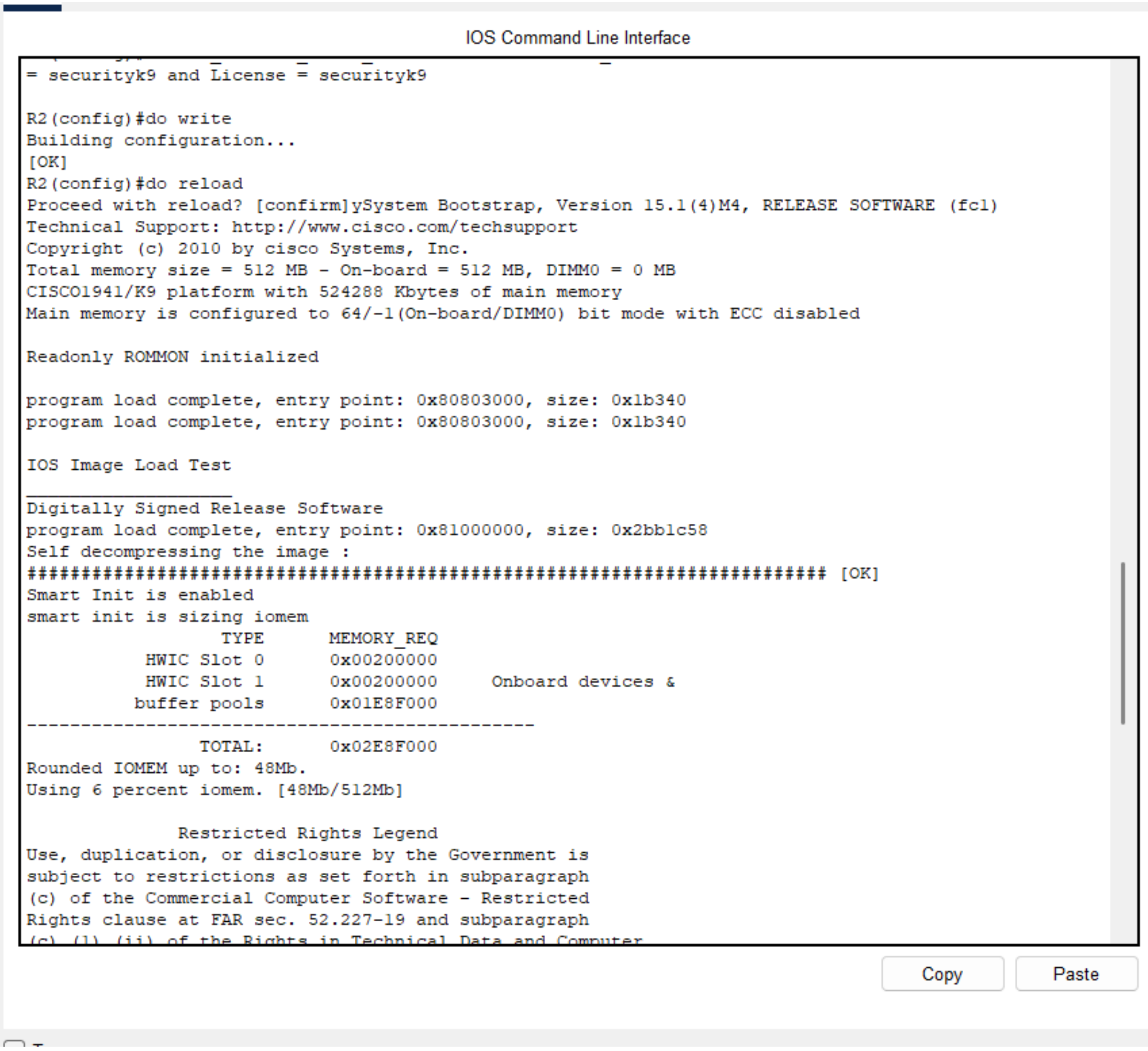




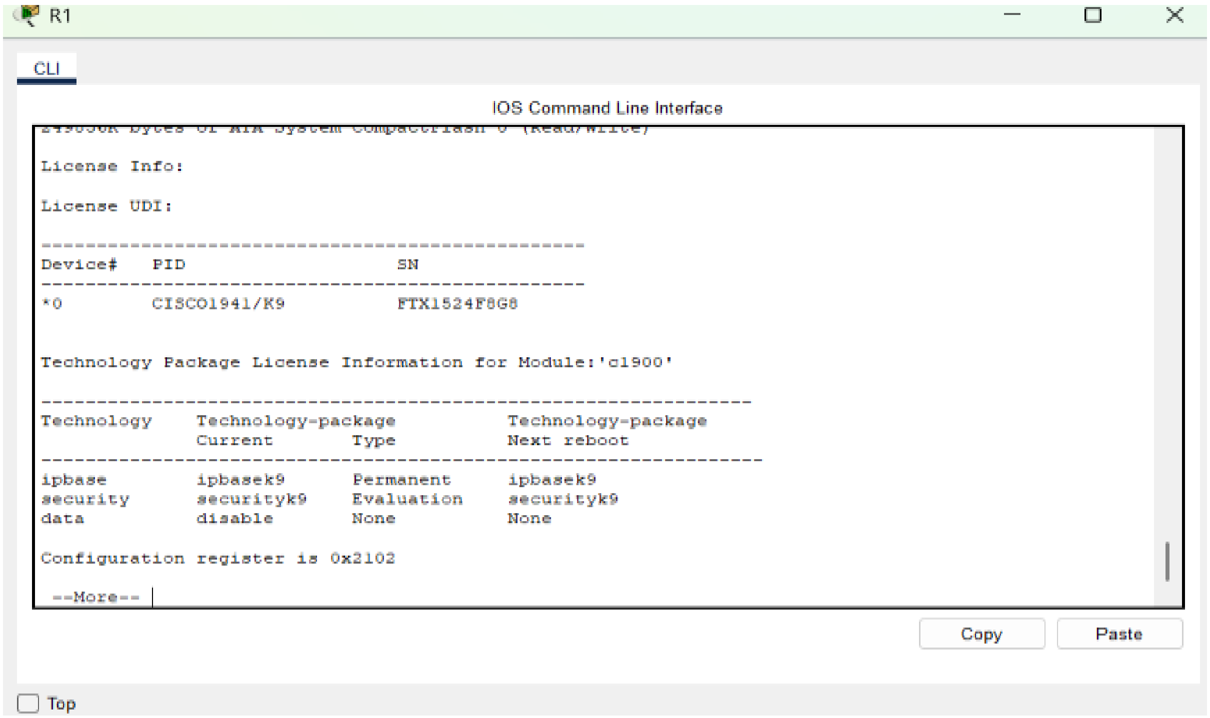
3. **Repeated the above steps on router R2**.

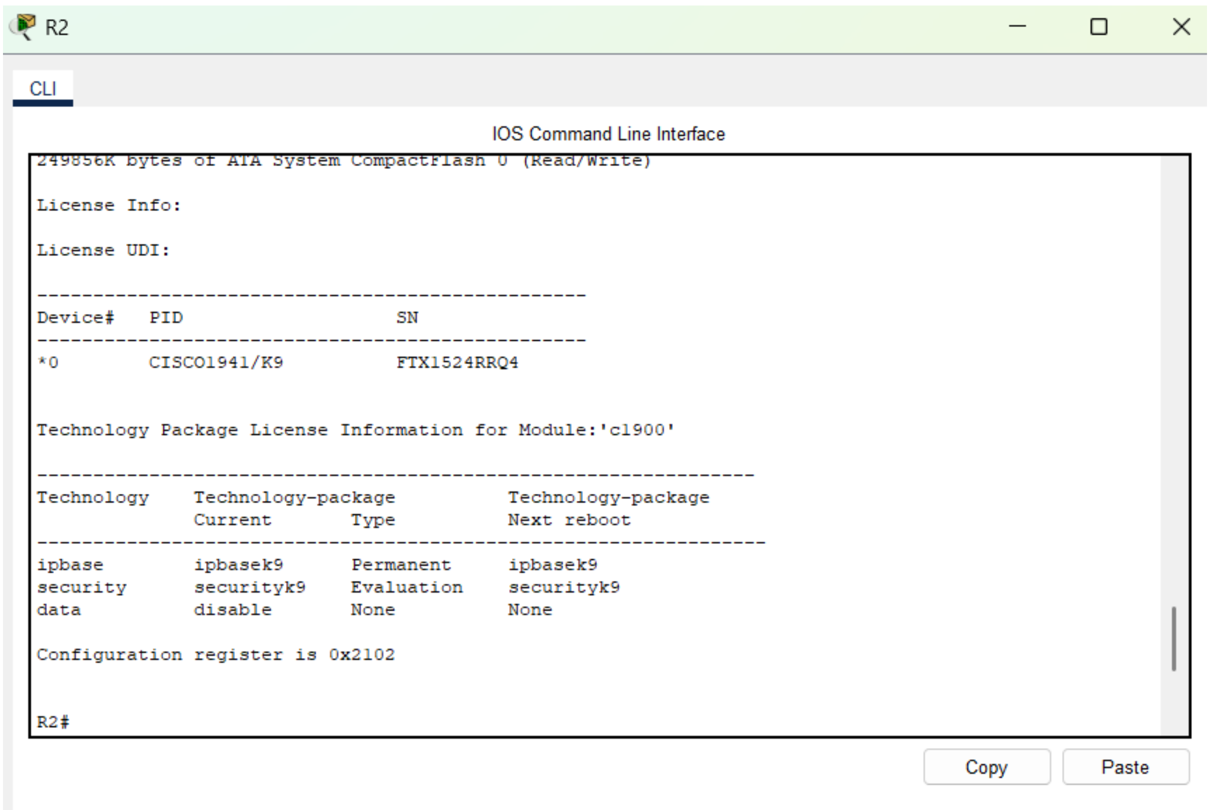



```
R2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#license boot module c1900 technology-package securityk9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
together with an additional  payment.  You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the  product,  including  during the 60 day  evaluation  period,  is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day  evaluation  period,  your  use of the  product  feature will be
governed  solely by the Cisco  end user license agreement (link above),
together  with any supplements  relating to such product  feature.  The
above  applies  even if the evaluation  license  is  not  automatically
terminated  and you do  not receive any notice of the expiration of the
evaluation  period.  It is your  responsibility  to  determine when the
evaluation  period is complete and you are required to make  payment to
Cisco for your use of the product feature beyond the evaluation period.

Your  acceptance  of  this agreement  for the software  features on one
product  shall be deemed  your  acceptance  with  respect  to all  such
software  on all Cisco  products  you purchase  which includes the same
software.  (The foregoing  notwithstanding, you must purchase a license
for each software  feature you use past the 60 days evaluation  period,
so  that  if you enable a software  feature on  1000  devices, you must
purchase 1000 licenses for use past  the 60 day evaluation period.)

Activation  of the  software command line interface will be evidence of
your acceptance of this agreement.


ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot
```



4. **After rebooting**, I confirmed activation using show version again and R1 and R2 were now activated.





In this section,

- The command `show version` is used to verify whether the security features (like IPsec) are enabled on the router. This is needed for VPN configuration.

• If security features are not active, the command `license boot module c1900 technology-package securityk9` enables them. This command tells the router to boot with the required security package.

• After making changes, the command `copy running-config startup-config` or `do write` saves the configuration so it's not lost after a reboot.

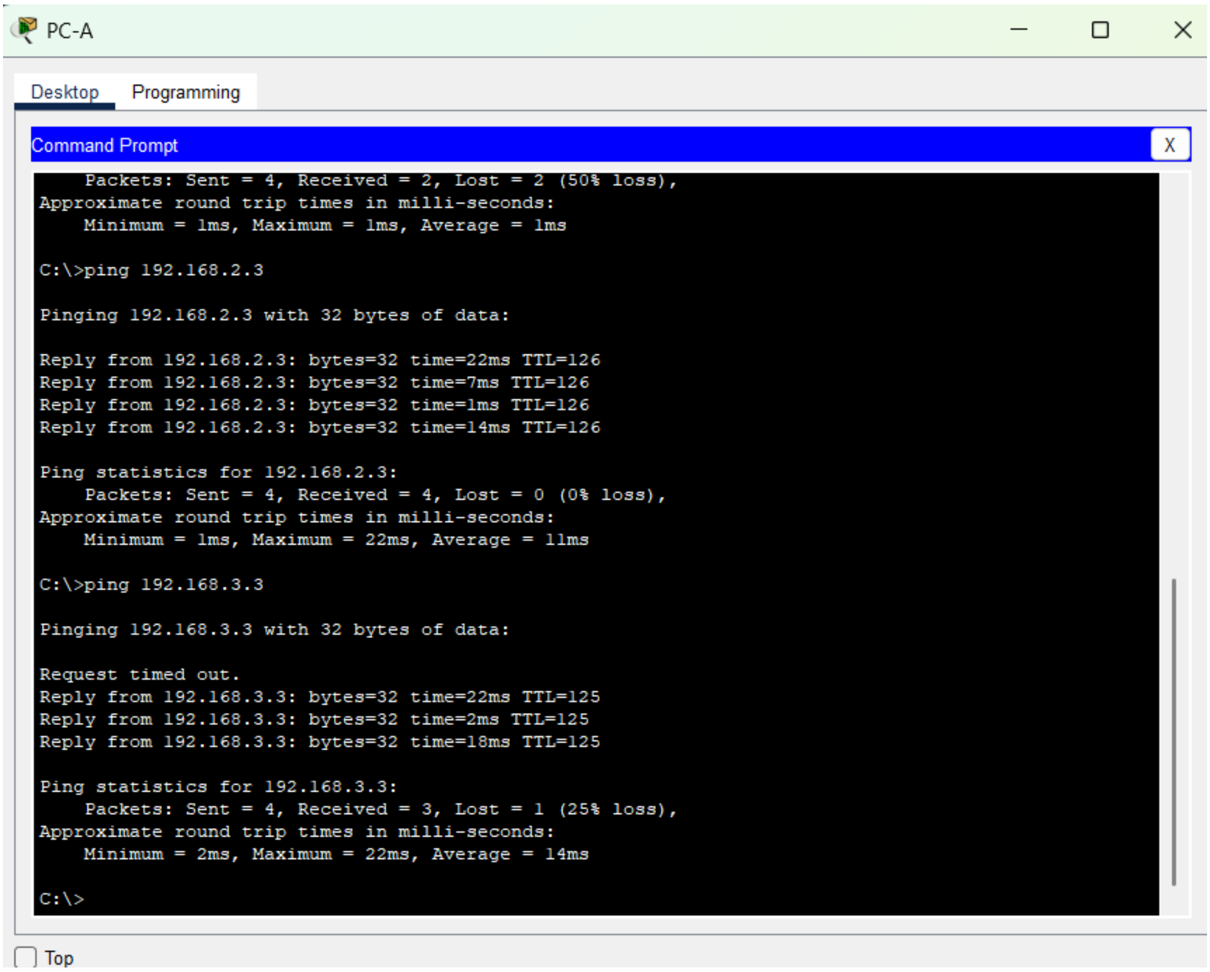
• The `reload` command restarts the router to activate the new license.

Part 2: Configure IPsec Parameters on R1

In this section, I learned how to configure IPsec VPN settings on R1 using access control lists, ISAKMP, and IPsec crypto maps.

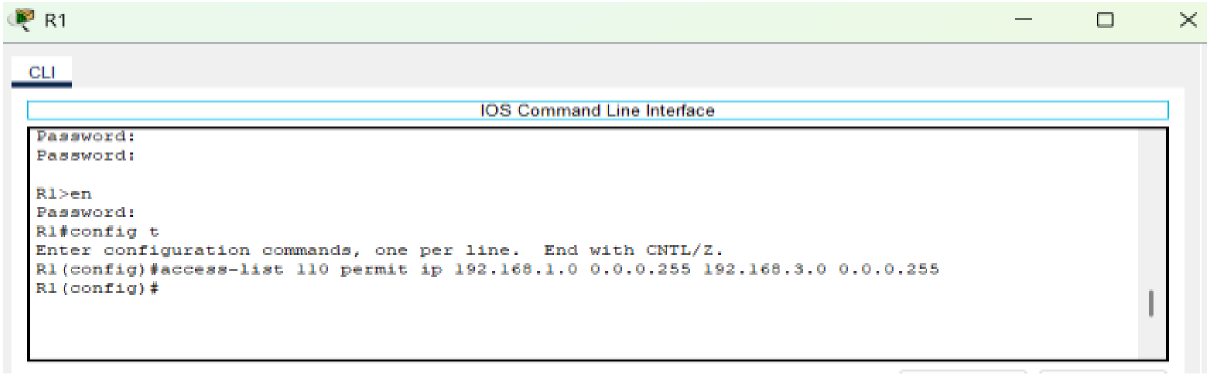
Steps followed:

1. **I Pinged PC-C from PC-A** to verify basic network connectivity.



2. **Created an ACL (110)** to define interesting traffic (LAN-to-LAN):

   R1(config)# access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255



3. **Configured ISAKMP Phase 1 on R1**:

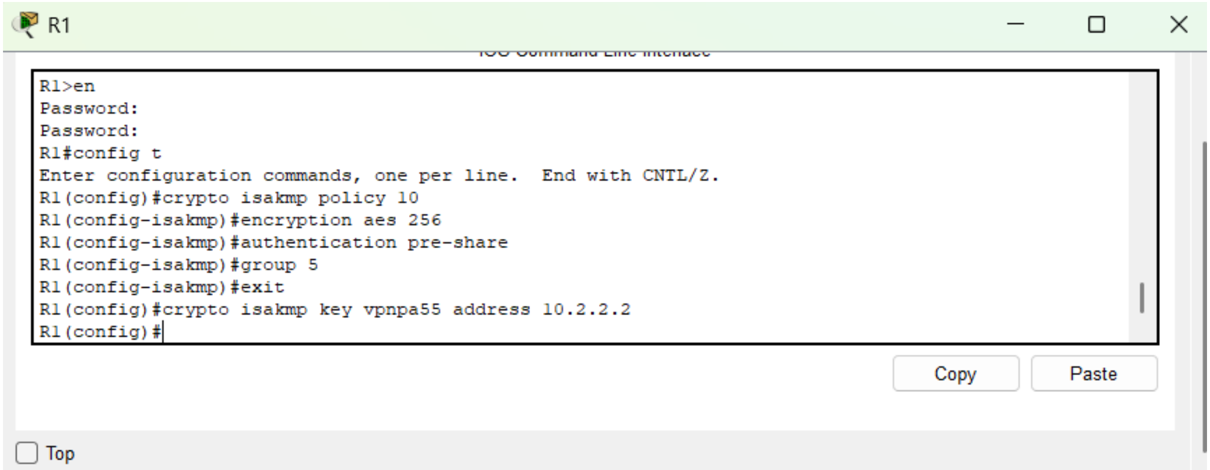
   R1(config)# crypto isakmp policy 10

   R1(config-isakmp)# encryption aes

   R1(config-isakmp)# authentication pre-share

   R1(config-isakmp)# group 2

   R1(config)# crypto isakmp key cisco address 10.2.2.2



4. **Configured IPsec Phase 2 (transform set and crypto map)**:

   R1(config)# crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac

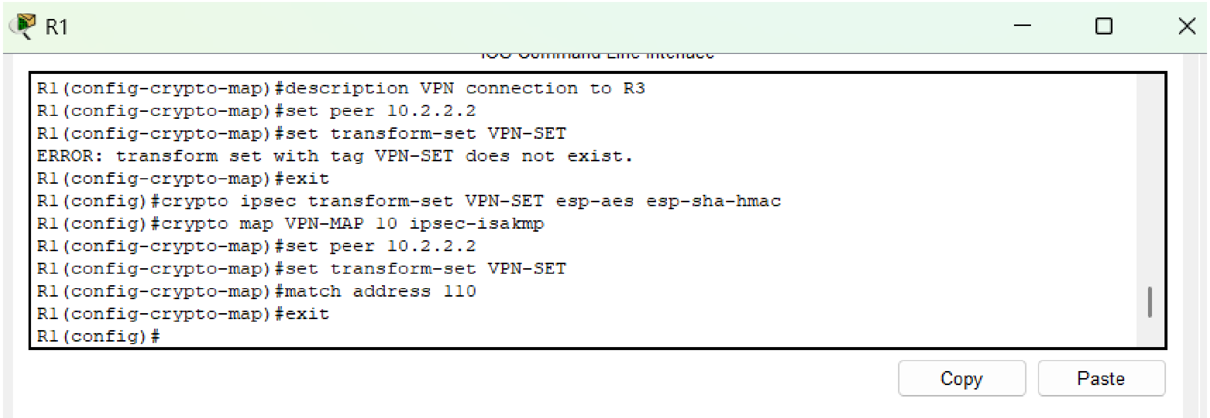
   R1(config)# crypto map VPN-MAP 10 ipsec-isakmp

   R1(config-crypto-map)# description VPN connection to R3

   R1(config-crypto-map)# set peer 10.2.2.2

   R1(config-crypto-map)# set transform-set VPN-SET

   R1(config-crypto-map)# match address 110

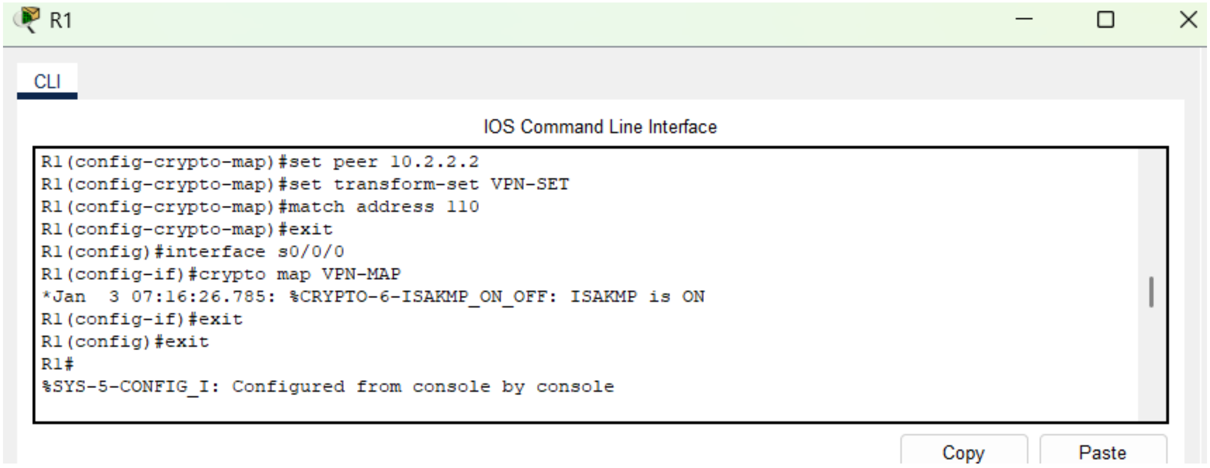


```
R1(config-crypto-map)#description VPN connection to R3
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
ERROR: transform set with tag VPN-SET does not exist.
R1(config-crypto-map)#exit
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#
```

5. **I Applied the crypto map to the outgoing interface**:

   R1(config)# interface s0/0/0

   R1(config-if)# crypto map VPN-MAP




```
R1(config-crypto-map)#set peer 10.2.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#exit
R1(config)#interface s0/0/0
R1(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

In summary in this section I did the following:

Step 1: Define Interesting Traffic

- The command `access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255` defines which traffic should be encrypted over the VPN. This is called "interesting traffic"—in this case, traffic from R1's LAN to R3's LAN.

Step 2: ISAKMP Phase 1 Configuration

- `crypto isakmp policy 10` creates a new IKE policy and assigns it a priority of 10. An IKE policy (Internet Key Exchange policy) is a set of rules that define how two VPN peers (like routers or firewalls) establish a secure tunnel between them—specifically for Phase 1 of the IPsec VPN negotiation.
- `encryption aes` sets the encryption algorithm to AES for securing Phase 1 traffic.
- `authentication pre-share` configures the use of a pre-shared key for authentication.

- `group 2` sets the Diffie-Hellman group for key exchange to group 2, which provides a balance of security and speed.
- `crypto isakmp key cisco address 10.2.2.2` sets the shared secret key (in this case, "cisco") and defines the IP address of the remote peer (R3).

Step 3: Configure IPsec Phase 2 (Transform Set and Crypto Map)

- `crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac` defines the encryption (3DES) and hashing (SHA) algorithms for encrypting the data.
- `crypto map VPN-MAP 10 ipsec-isakmp` starts the creation of a crypto map and links it to ISAKMP.
- `set peer 10.2.2.2` tells the router who the VPN peer is (R3's serial IP).
- `set transform-set VPN-SET` links the earlier encryption settings to this VPN.
- `match address 110` binds the access list (the defined interesting traffic) to the crypto map.

Step 4: Apply the Crypto Map to the Interface

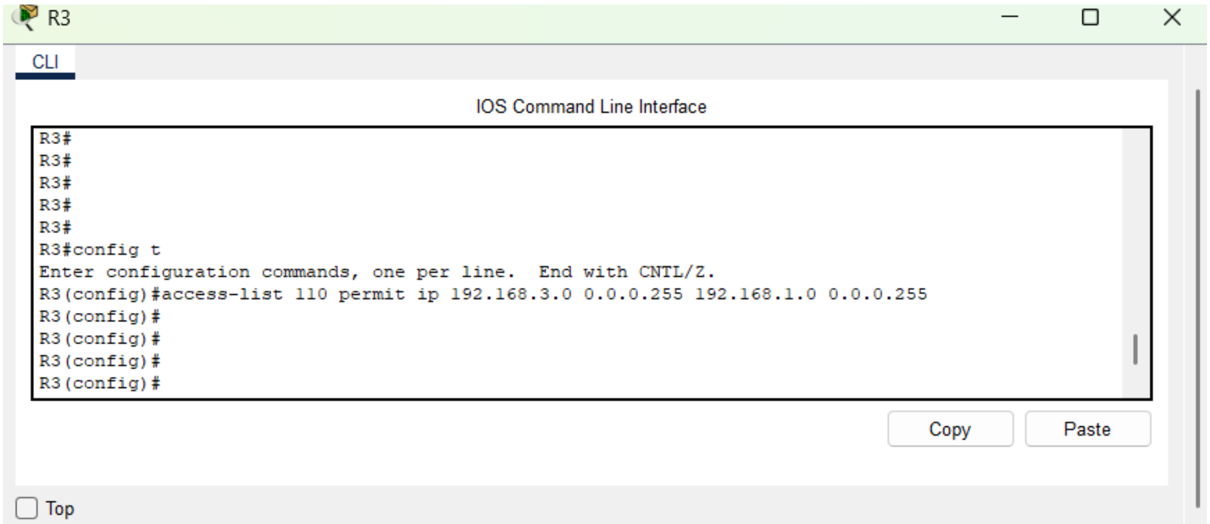
- `interface s0/0/0` enters the serial interface configuration mode.
- `crypto map VPN-MAP` applies the VPN configuration to this physical interface so the router knows to secure the traffic that passes through it.

Part 3: Configure IPsec Parameters on R3

In this section, I learned to configure the same IPsec VPN settings on R3 to establish a site-to-site tunnel with R1.

1. **Created ACL (110)** for interesting traffic from R3 to R1:

   R3(config)# access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255

2. **Configured ISAKMP Phase 1 on R3**:

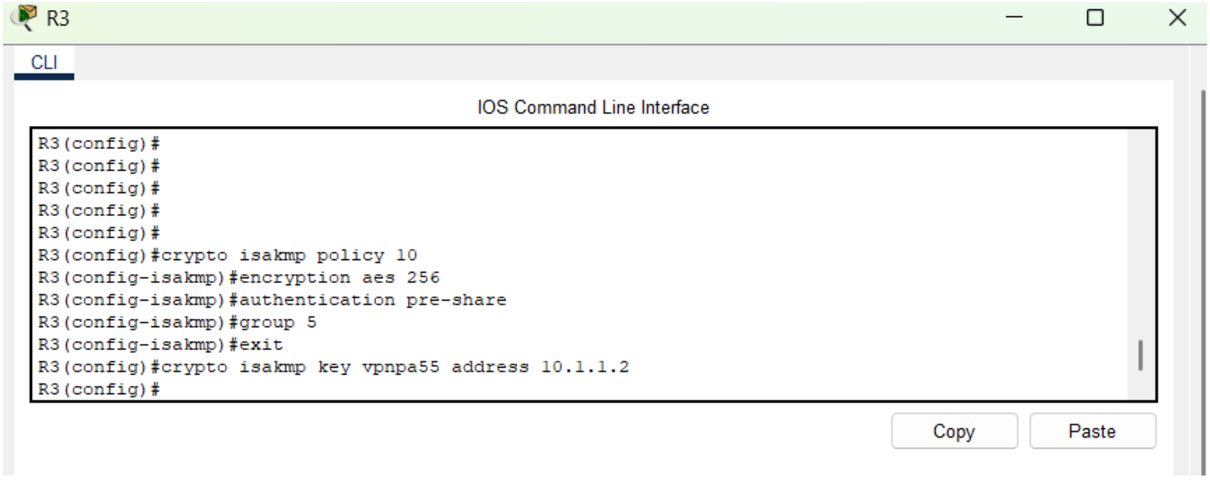
R3(config)# crypto isakmp policy 10

R3(config-isakmp)# encryption aes

R3(config-isakmp)# authentication pre-share

R3(config-isakmp)# group 2

R3(config)# crypto isakmp key cisco address 10.1.1.2



3. **Configured IPsec Phase 2 (transform set and crypto map)**:

   R3(config)# crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac

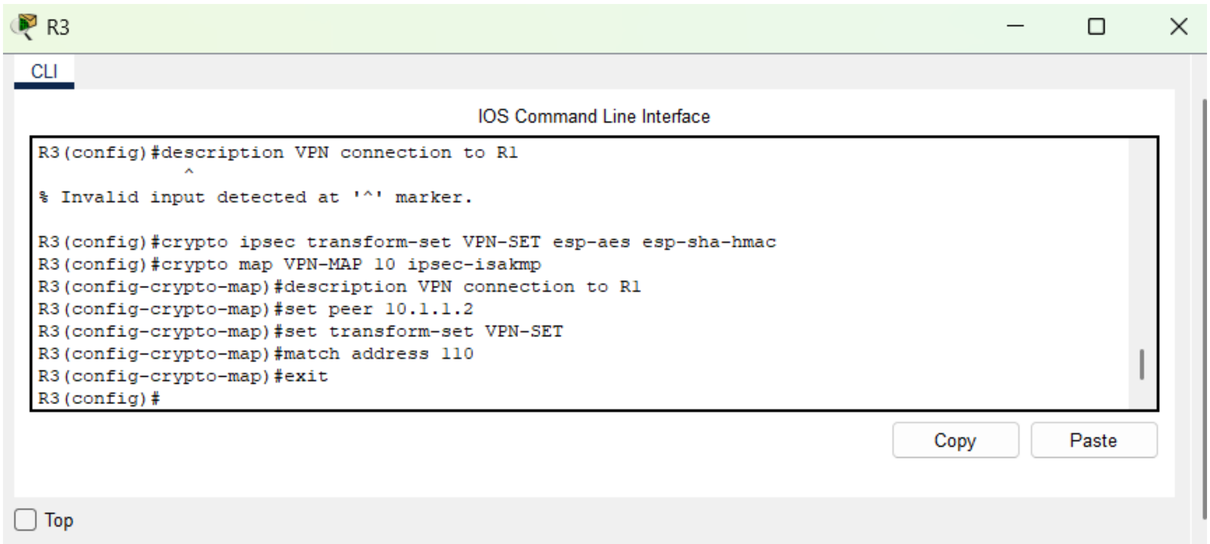
   R3(config)# crypto map VPN-MAP 10 ipsec-isakmp

   R3(config-crypto-map)# description VPN connection to R1

   R3(config-crypto-map)# set peer 10.1.1.2

   R3(config-crypto-map)# set transform-set VPN-SET

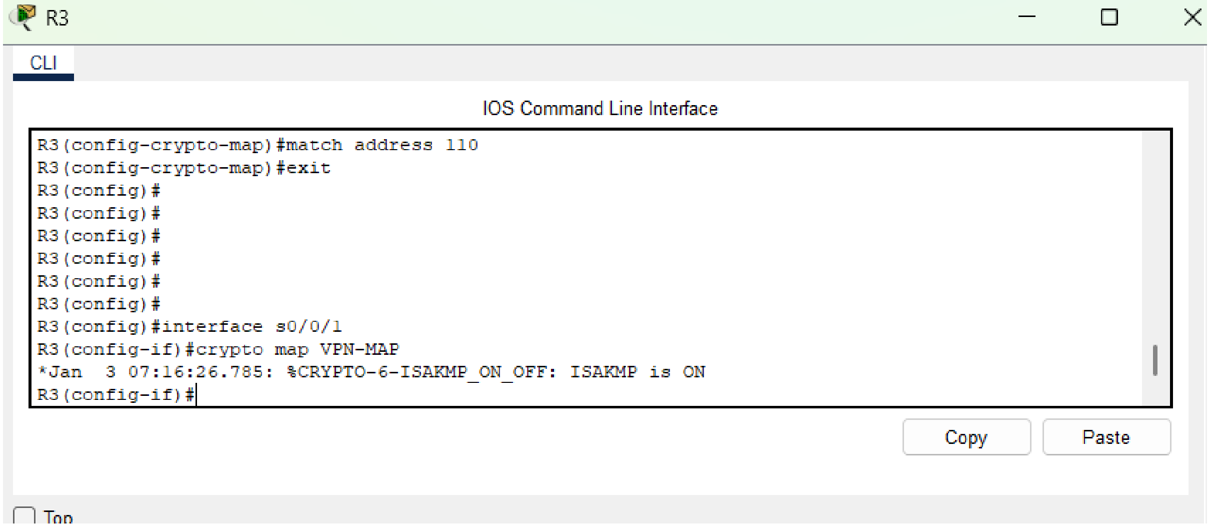
   R3(config-crypto-map)# match address 110



4. **Applied the crypto map to outgoing interface s0/0/1**:

   R3(config)# interface s0/0/1

   R3(config-if)# crypto map VPN-MAP



In Summary, this section covered the following:

Step 1: Define Interesting Traffic

- `access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255` marks the traffic coming from R3's LAN and going to R1's LAN as interesting.

Step 2: ISAKMP Phase 1 Configuration

- `crypto isakmp policy 10` creates a matching IKE policy.
- `encryption aes` sets AES as the encryption algorithm.
- `authentication pre-share` configures the same shared key method.
- `group 2` sets the key exchange method.
- `crypto isakmp key cisco address 10.1.1.2` specifies the same pre-shared key and points to the R1 peer IP.

Step 3: IPsec Phase 2 Configuration

- `crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac` defines the encryption and integrity mechanisms.
- `crypto map VPN-MAP 10 ipsec-isakmp` creates the crypto map container.
- `set peer 10.1.1.2` sets the VPN peer (R1's IP).
- `set transform-set VPN-SET` connects the crypto map to the transform set.
- `match address 110` binds the ACL to the VPN.

Step 4: Apply to Interface

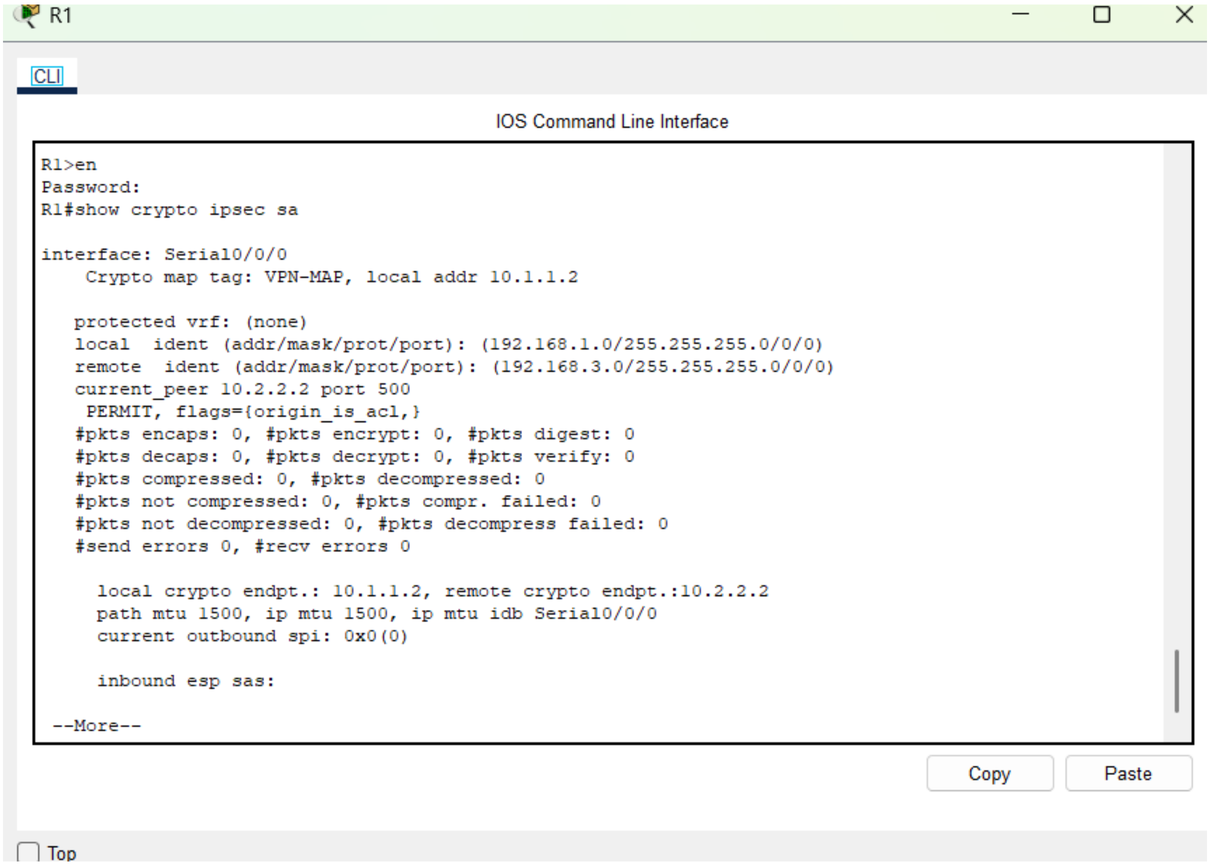
- `interface s0/0/1` accesses R3's serial interface.
- `crypto map VPN-MAP` activates VPN settings on that interface.

Part 4: Verify the IPsec VPN

In this section, I learned how to verify VPN tunnel establishment and observe packet encryption behavior.

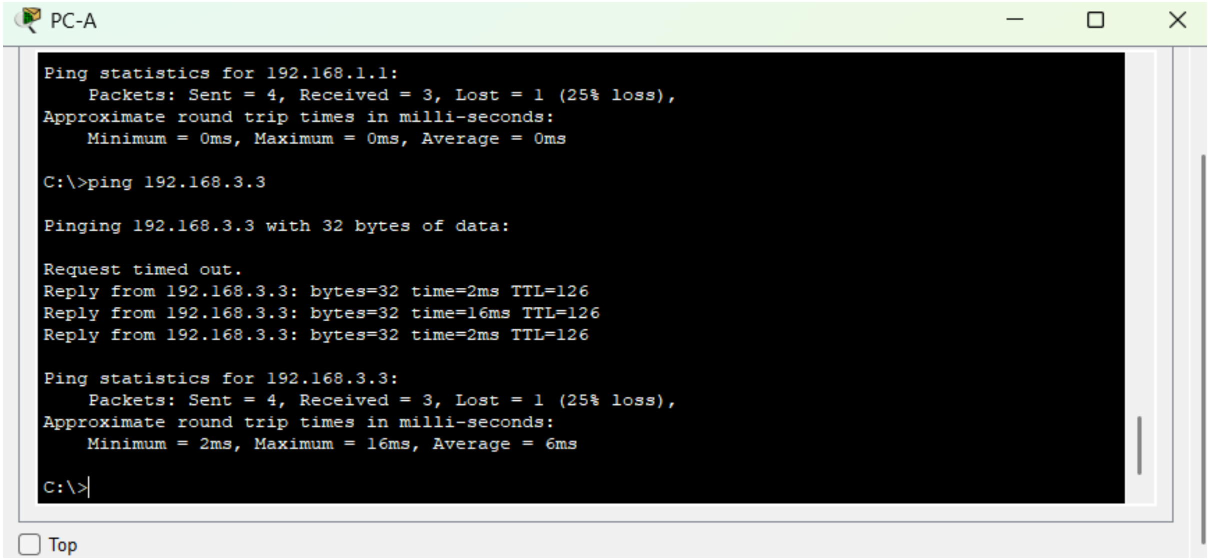
1. **Checked IPsec status before traffic**:

   R1# show crypto ipsec sa



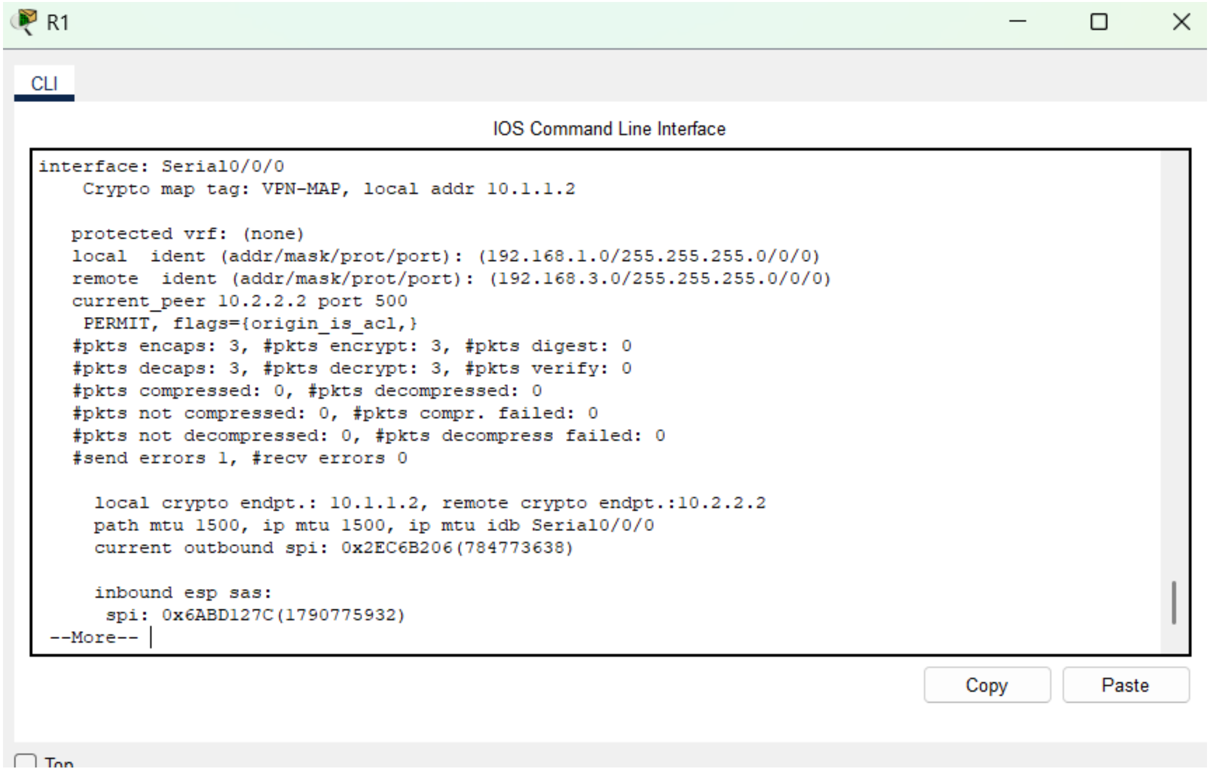
→I Verified that no packets were yet encrypted (all counters at 0).

2. **Created interesting traffic** by pinging PC-C from PC-A.



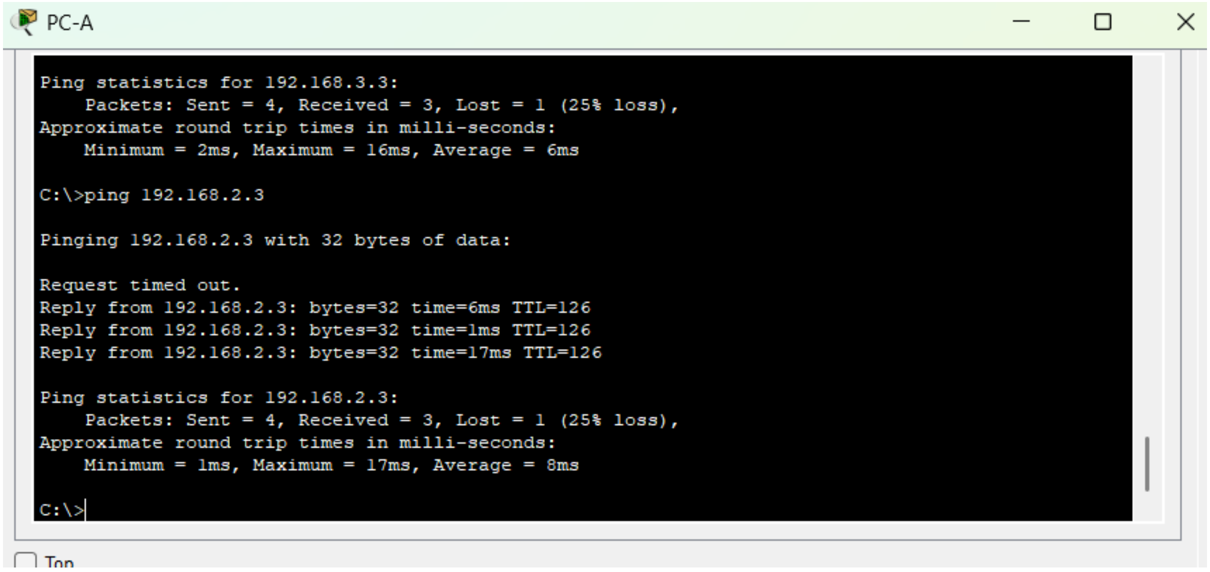
3. **Re-checked IPsec status**:

   R1# show crypto ipsec sa

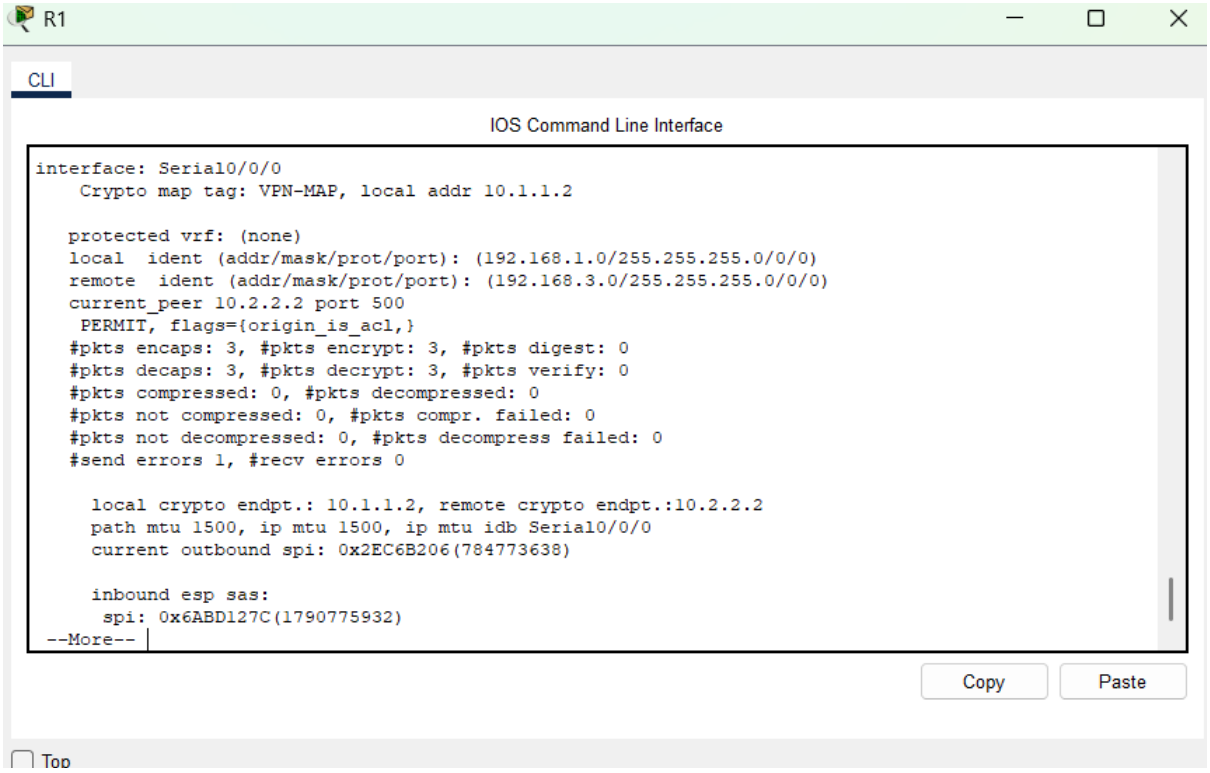


→I hence Confirmed that packet counters incremented (VPN tunnel active).

4. **Created uninteresting traffic** by pinging PC-B from PC-A to ensure no VPN was triggered.



The ping was successful, however there was no encryption as shown below:



The number of encrypted packets is still as it was on the previous ping no additions meaning no encryption was done on the uninteresting traffic.

In summary, in this section I used the following commands, and these were their purpose:

- `show crypto ipsec sa` shows IPsec Security Associations. You'll use this to see if packets are being encrypted and decrypted, which confirms the tunnel is working.

- `ping 192.168.3.x` from PC-A initiates the interesting traffic to trigger the VPN tunnel.

- After the ping, running `show crypto ipsec sa` again shows incremented packet counters, which proves that traffic is now going through the encrypted tunnel.

- `show crypto isakmp sa` checks whether the IKE Phase 1 negotiation was successful.

- `show crypto map` shows you the crypto map configuration applied to the interfaces

Conclusion

By completing this lab, I gained practical experience in configuring and verifying an IPsec VPN tunnel between two routers. I learned how to apply security policies and how VPN tunnels are activated based on interesting traffic. The activity demonstrated how secure communication can be established across a public network while ensuring data integrity and confidentiality. This exercise solidified my understanding of VPN principles and router-based security configuration using Cisco technologies